

Luckie Musngi

(520) 999-5482 | luckie.musngi@gmail.com | <https://github.com/LuckieMusngi/>

EDUCATION

University of Arizona (Graduated May 2026)

Tucson, AZ

- **Bachelor of Science** in Computer Science, **Minor** in Mathematics
- **Relevant Coursework:** Operating Systems, Database Design, Computer Networks, Web Programming

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C/C++

Backend & Tools: Node.js, Express.js, PostgreSQL, MongoDB, Git

Data & ML: Pandas, NumPy, Scikit-learn, Matplotlib

EXPERIENCE

University of Arizona

Tucson, AZ

Undergraduate Researcher, Bio Robotics & Scientific Instrumentation

Jan 2026 - May 2026

- Designed and evaluated automated robotic systems for experimental workflows in chemistry environments
- Integrated sensors, actuators, and control systems to support repeatable automated procedures
- Performed system testing and reliability evaluation through iterative experimental validation

Lin's Grand Buffet

Tucson, AZ

Expeditor

July 2022 - Present

- Coordinated high-volume workflow between kitchen stations under time constraints
- Communicated across teams to resolve operational bottlenecks during peak service hours
- Maintained efficiency and accuracy in fast-paced, real-time environment

Vail Academy and High School

Tucson, AZ

Site Tech Coordinator Shadow

Sep 2021 - Nov 2021

- Assisted with troubleshooting hardware, software, and network issues
- Supported technical setup for school events and classroom systems
- Gained exposure to IT support workflows and infrastructure maintenance

PROJECTS

Tucson Crime Analytics & Classification System | <https://github.com/LuckieMusngi/Crime-in-Tucson-Analysis>

- Built end-to-end data processing pipeline for large-scale crime datasets including cleaning, feature engineering, and transformation
- Developed and evaluated classification models using cross-validation and performance metrics (precision, recall, F1-score)
- Engineered structured dataset outputs enabling statistical trend analysis and predictive modeling
- Visualized crime patterns and model performance for interpretability and reporting

Tech: Python | Pandas, NumPy, Scikit-learn, Matplotlib

Environmental Health Data ETL + PostgreSQL System |  Air vs. Health RDB

- Designed normalized relational schema integrating messy and unrelated environmental and health datasets
- Built ETL pipeline processing 100K+ records with structured cleaning and transformation logic
- Implemented bulk data ingestion into PostgreSQL using optimized insert operations
- Performed relational queries to analyze correlations between environmental and health indicators

Python | PostgreSQL, psycopg2, Pandas

Ski Resort Relational Database System | <https://github.com/LuckieMusngi/Ski-Resort-Database>

- Designed normalized relational database schema with constraints and entity relationships
- Implemented CRUD operations using JDBC and SQL query integration
- Built command-line interface for database interaction and management
- Applied ER modeling principles to structure relational data system

Java | SQL, JDBC

Bank Website Management System | <https://github.com/LuckieMusngi/BankWebsite>

- Built full-stack web application supporting user authentication and session-based workflows
- Developed RESTful backend APIs using Express.js
- Integrated MongoDB for persistent data storage and retrieval
- Implemented multi-page user flow and state management

Node.js | Express, MongoDB

High-Speed Arduino Sorting Machine | <https://github.com/LuckieMusngi/Arduino-Skittle-Sorter> [Quannel-Method](#)

- Designed and implemented high-speed embedded sorting system with real-time control logic
- Integrated sensors and actuators for automated object classification and routing
- Engineered mechanical system using CAD-designed components for optimized flow
- Achieved throughput of ~4.4 items/sec through timing and control loop optimization

Arduino | C/C++, OpenSCAD

Real-Time Procedurally Generated Game (Unity / C#) | <https://github.com/LuckieMusngi/Unity-Roguelike>

- Built procedural level generation system for replayable gameplay environments
- Implemented AI behavior systems and real-time combat mechanics
- Designed modular architecture using object-oriented Unity components
- Used coroutine-based event sequencing for runtime game flow control

C# | Unity